DATA MINING & MACHINE LEARNING

# FakeNews Detector

Data Mining & Machine Learning Project
M.Sc. in Artificial Intelligence and Data Engineering — Academic Year 2025–2026

Alessio Grillo — ID: 654947 | Andrea Giacomazzi — ID: 720164

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OVERVIEW

# Presentation Outline

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## Problem & Dataset
Task definition · features · EDA

02
## Pipeline & Experiments
Preprocessing · model selection · hyperparameter optimization

PART 1

03
## Evaluation & Results
Metrics · cross-validation · final model performance

04
## Explainability
Global & local techniques · consistency with domain knowledge

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Error analysis · limitations · future directions

PART 1 — TECHNICAL

# Problem, Pipeline & Results

Problem definition · feature engineering · model selection
hyperparameter optimization · evaluation · test-set results

PROBLEM

# Problem Definition & Dataset

## The Problem

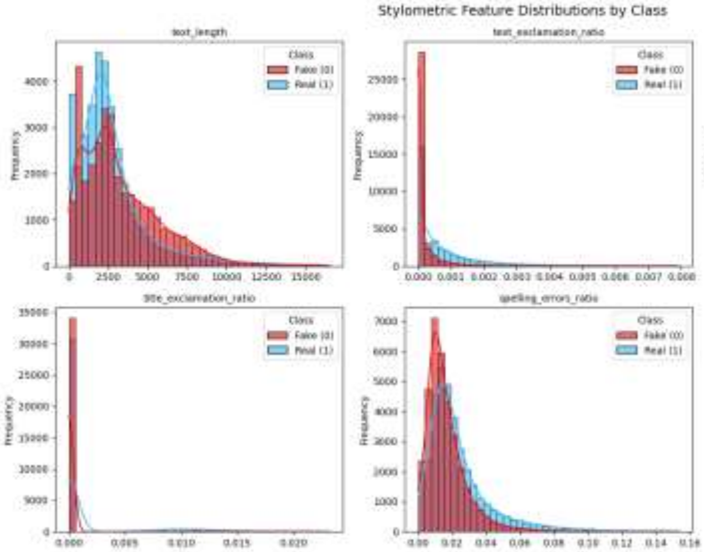
- Given a **news article**, predict whether it is credible or deceptive based on its text alone.
- Task type: binary classification
- Fake news spreads faster than anyone can fact-check it, shaping opinions on elections, health, and public trust.
- A model cannot check whether something is true facts, it can only detect stylistic and linguistic patterns. And the best fake articles are written to sound exactly like real ones.

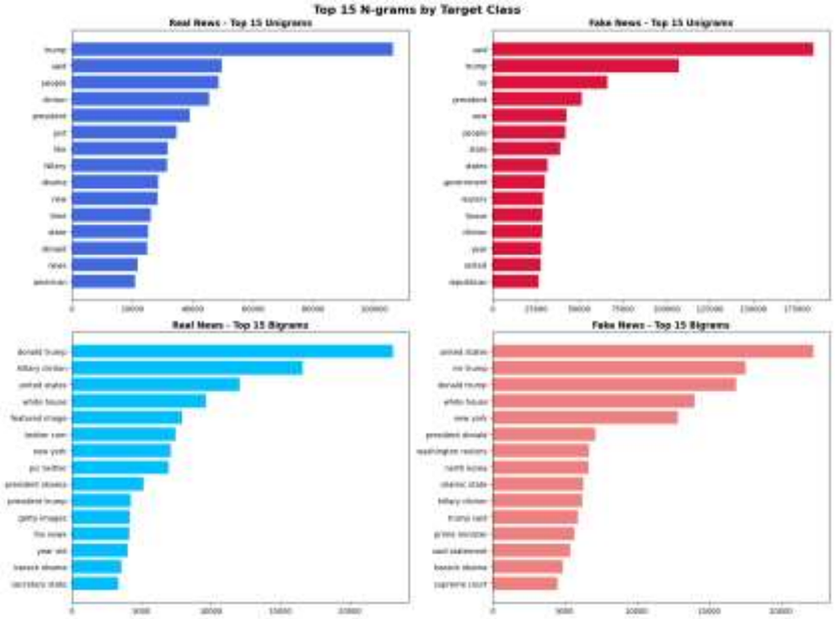
## Dataset at a Glance

| | |
|---|---|
| **Task** | *Binary classification* |
| **Dataset** | *Kaggle, 72.134 samples, 3 features* |
| **Target** | *Article veracity (0: Fake News, 1: Real News)* |
| **Features** | *1 numerical, 2 textual* |
| **Missing values** | *0,21% overall* |
| **Class balance** | *51.5% / 48.5% balanced* |
| **Split** | *80% Training / 20% Testing* |
| **Metric(s)** | *Accuracy, F1-score, Precision, Recall, AUC-ROC* |

PROBLEM

# Exploratory Data Analysis




### Key findings from EDA:

**Stylometric features** reveal significant structural and stylistic differences between classes, proving that how a news article is written is highly indicative of its veracity.
**N-gram** The N-gram analysis reveals strong dataset-specific biases rather than purely semantic differences.

PIPELINE

# Feature Engineering & Selection

## Engineering

- Text vectorization using TF-IDF (Unigrams & Bigrams)
- 10,000 TF-IDF features from concatenated title + body.
- Extraction of 5 stylometric features: title uppercase ratio, text length, spelling error rate, exclamation & question mark density (both title and text).
- MinMax scaling on the 5 stylometric features to align their magnitude with the sparse TF-IDF matrix.
- Concatenated the sparse TF-IDF matrix with the dense, scaled stylometric features to create a unified feature space.

## Selection Method

- Frequency-based Filtering: Limited the TF-IDF vocabulary to the top 10,000 most frequent tokens to reduce dimensionality and eliminate rare noise/typos (we discarded some stopwords based on the n-gram plot).
- Evaluated the unified feature pool using a Chi-Square test.
- Title uppercase ratio emerged as the first overall predictor, and title exclamation ratio ranked in the Top 20.
- Retained features with lower Chi-Square scores (text length, exclamation ratio, spelling errors) because EDA shows they contain critical structural anomalies. This preserves domain robustness against keyword overfitting.

## Why These Features?

- TF-IDF captures vocabulary shifts and source-mimicking artifacts confirmed by EDA.
- Stylometric features target sensationalism signals that differ significantly between classes.

PIPELINE

# ML Pipeline: Avoiding Data Leakage & Bias

Raw Data → Preprocessing → Feature Selection → Model Training → Validation (held-out fold) → Final Evaluation

### Data Leakage Prevention

- All preprocessing steps (TF-IDF, MinMax scaler, stopword removal) wrapped in a Scikit-Learn Pipeline, fitted exclusively on training folds.
- Train/test split performed before any transformation. Test folds only ever see frozen parameters.
- No test set distribution ever influences the vocabulary, scaling bounds, or stopword list.

### Class Imbalance Handling

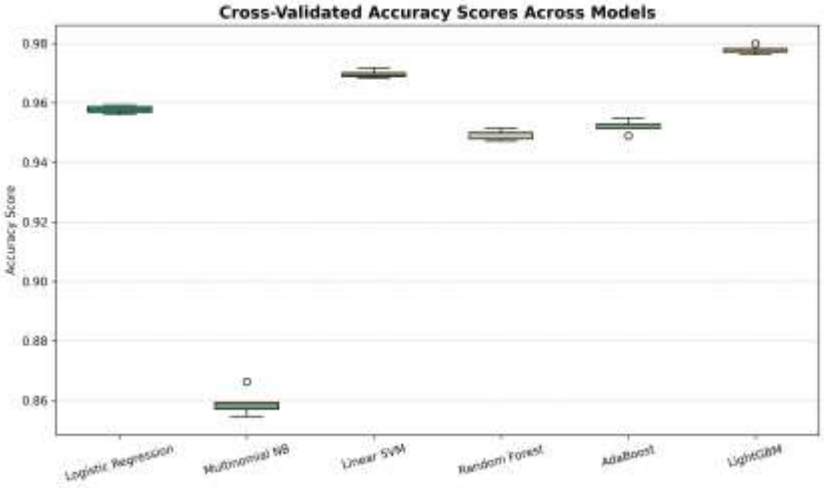
- No balancing technique applied. The dataset is naturally balanced (51.5% fake / 48.5% real), so SMOTE and class weights were unnecessary.
- Stratified 5-fold CV preserves the natural class ratio across every split, ensuring stable and comparable metrics throughout.

### Metric Justification

- **Primary**: **F1-score & Accuracy**. Balanced classes make accuracy meaningful; F1 jointly captures precision and recall.
- AUC-ROC reported as secondary metric to assess discriminative power across all decision thresholds.

PIPELINE

# Model Selection & Hyperparameter Optimization

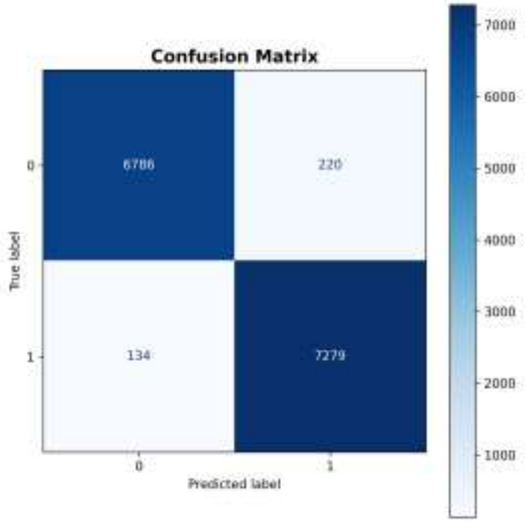


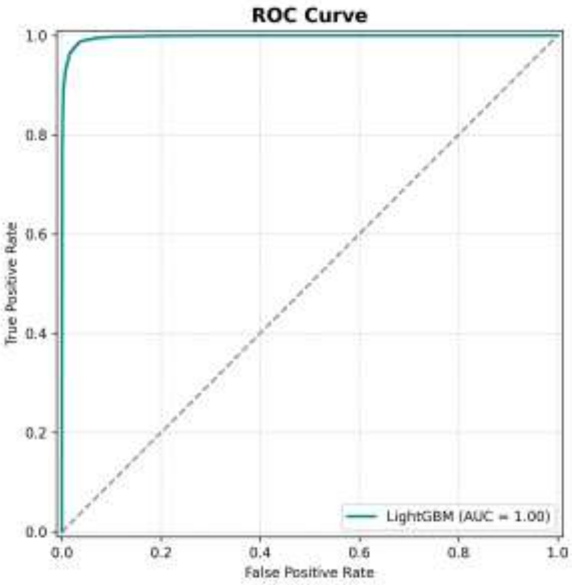

## Key Decisions

| | |
|---|---|
| **Candidate models** | *Linear baseline (LR, NB, SVM) + ensemble (RF, AdaBoost, LightGBM) covering linear, probabilistic, geometric and boosting families* |
| **CV strategy** | *Stratified 5-fold CV for all models* |
| **Selection criterion** | *Accuracy* |
| **Chosen model** | *LightGBM: highest CV accuracy (0.978) with lowest variance* |
| **Search strategy** | *Random Search (5 combinations x model)* |
| **Search space** | *n_estimators: {50,100,150} · learning_rate: {0.05, 0.1} · num_leaves: {15, 31}* |
| **Best config** | *n_estimators: 100 · learning_rate: 0.1 · num_leaves: 31* |
| **Statistical test** | *McNemar's test: LightGBM vs Linear SVM · $\chi^2$=20.32 · p=0.00001, $\alpha$=0.05* |

RESULTS

# Test-Set Results




### Key Metrics (Test Set)

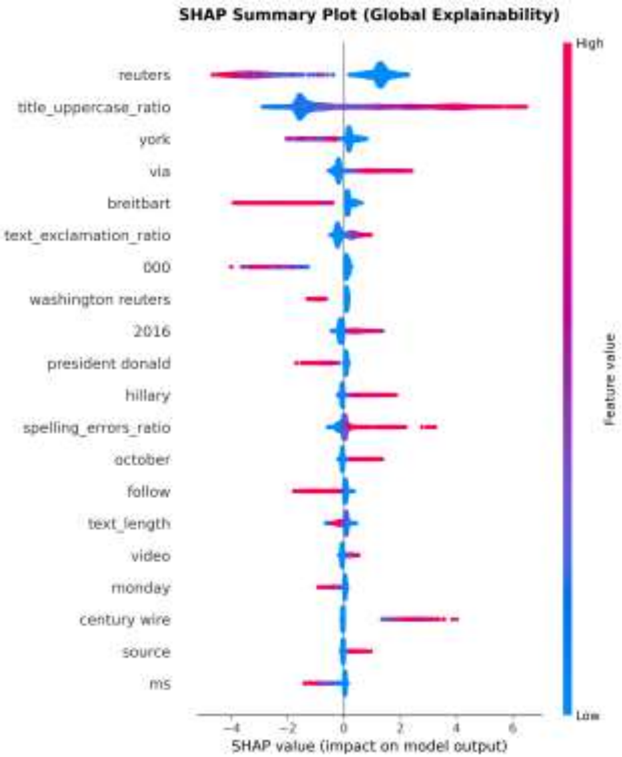
| | | |
|---|---|---|
| **97,5%** | **97,6%** | **0,997** |
| Accuracy | F1-Score | AUC-ROC |

PART 2 — CRITICAL ANALYSIS

# Explainability & Critical Discussion

How does the model work? · What did it get wrong and why?
Limitations · future directions · domain impact

# Global Explainability: What Drives the Model Overall?

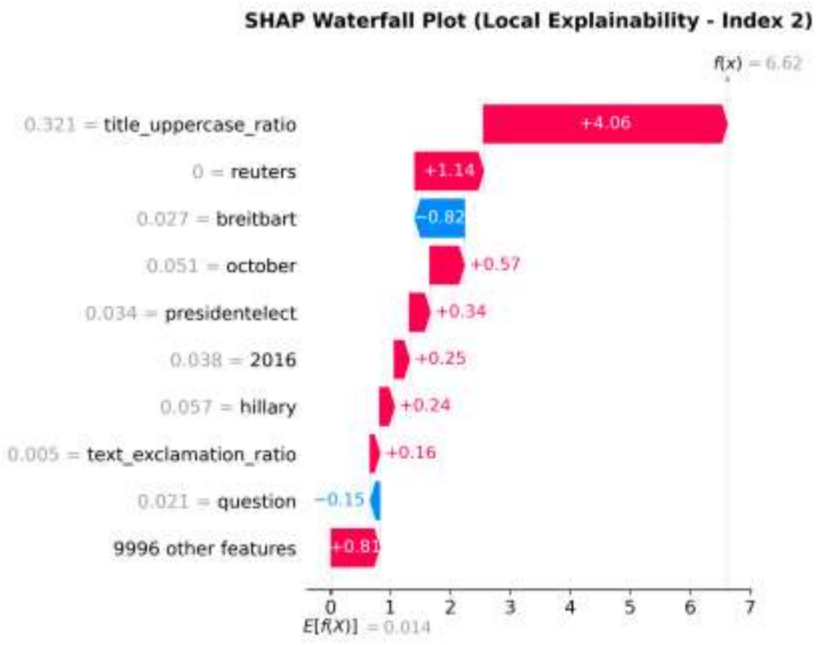



## Interpretation

- **SHAP beeswarm analysis**:
  - Each dot = one sample
  - Color = feature value
  - X-axis = feature impact
- **Key signals**:
  - Presence of *Breitbart* -> pushes prediction toward Fake
  - Absence of *Reuters* -> associated with Real
- **Insight**:
  - Model uses meaningful source patterns (not noise)
  - Suggests learned distinction between **partisan** and **institutional sources**
  - High accuracy (97.5%) suggests the model is learning meaningful signals

EXPLAINABILITY

# Local Explainability: Why This Specific Prediction?

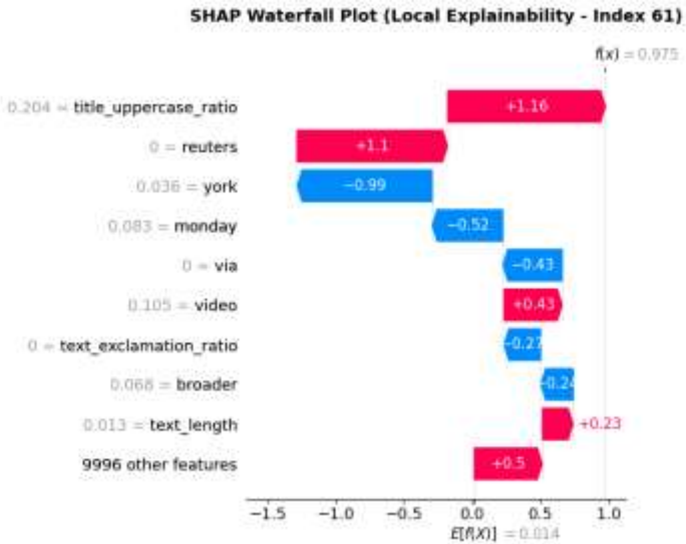



## Case Analysis

- **Sample**: Article from October 2016 mentioning "president-elect", "Hillary", and "2016" — political content from the US election period, ground truth: True News (correctly classified)
- **Key drivers**: *title_uppercase_ratio* (+4.06) dominates; *reuters* absent (value=0, +1.14) means the absence of this trusted token pushes toward real; *breitbart* presence drives toward fake as expected
- **Local & Global alignment**: top features here (*title_uppercase_ratio*, *reuters*, *breitbart*) match the global SHAP ranking — consistent behavior across instances.
- **Conclusion**: An article with high capitalization, no press agency signature, and partisan political references make the model reasons correctly.

CRITICAL DISCUSSION

# Error Analysis: What Did the Model Get Wrong and Why?




## Reverse-Engineering Errors

- **Two failure modes**: FN (134): fake news mimicking legitimate reporting; FP (220): genuine opinion content penalized for aggressive style
- **Root cause**: the model confounds writing style with factual content, stylometric proxies cannot capture semantic intent
- Index 61 (FP): *title_uppercase_ratio* and *reuters* absent push toward Real, but *york* and *monday* pull back.
- Errors are **systematic**, not random: misclassifications cluster around the stylometric decision boundary, confirming a known architectural limit of TF-IDF pipelines

## Comparison with Prior Work

**Verma** et al. (2021) — the original WELFake paper — reports ~**95% accuracy** using word embeddings combined with 20 hand-crafted linguistic features. Our LightGBM pipeline achieves **97.9%** with a simpler approach: TF-IDF + only 5 stylometric features. We outperform the dataset authors' own benchmark while maintaining full interpretability via SHAP.

CRITICAL DISCUSSION

# Domain Impact, Limitations & Future Directions

## Domain Impact

- First-pass filter for content moderation, flags suspicious articles before publication or amplification.
- Supports fact-checkers by prioritizing review queues, not replacing human judgment.
- Balanced error rate (FP censors journalism; FN lets misinformation spread) makes it a decision-support tool, not a final arbiter.
- SHAP explanations make predictions auditable, essential for human-facing moderation.

## Limitations

- WELFake is English-only and US-politically skewed, generalization untested.
- Static TF-IDF is vulnerable to concept drift; performance degrades without retraining.
- The model detects stylistic proxies of deception, not factual truth.
- Evaluated on a single dataset split, no real-world feedback loop.

## Future Directions

- Replace TF-IDF with contextual embeddings (BERT) for deeper semantic understanding.
- Integrate fact-checking APIs to cross-reference claims against verified databases.
- Extend to multimodal input (images + metadata) to catch deception beyond text.
- Deploy with human-in-the-loop feedback to measure calibration and detect drift.

# Thank You

**Conclusions:**

- Problem: automated binary classification of news articles to combat digital misinformation, using semantic and stylometric features
- Best model: LightGBM with 97.9% accuracy and F1, validated via Stratified 5-Fold CV and confirmed statistically superior to Linear SVM (McNemar's, $p=0.00016$)
- Key XAI finding: SHAP reveals the model learned journalistically meaningful signals, source tokens (breitbart, reuters) and sensationalist stylometry; not dataset artifacts, confirming genuine generalization
- Main limitation & next step: TF-IDF cannot distinguish writing style from factual content; future work should explore transformer-based architectures (BERT, RoBERTa) for deep contextual understanding

Alessio Grillo · a.grillo6@studenti.unipi.it
Andrea Giacomazzi · a.giacomazzi1@studenti.unipi.it